

20 per cent of profits, and the application field 10 to 20 per cent."

In 2007, the total global LED market exceeded \$6 billion—up by 13.7 per cent from the previous year. During 2006 to 2009, the annual compound growth rate of the global LED market was 10 per cent, most of which was mainly contributed by the ultra-high- and high-brightness LEDs. The growth rate went up significantly in 2011 as prices fell, thanks to increased production in China.

According to research firm LEDinside, the LED industry made about 100 billion LED chips in 2011 for a market that needs only 89 billion. That 12 per cent 'oversupply ratio' will worsen to 21 per cent in 2012 after LED makers add manufacturing capacity. This would mean further drop in prices of LED bulbs and tubes.

The global LED industry is mainly concentrated in Japan, Taiwan, Europe, America, South Korea and China. Japan, the largest producer in the global LED industry, holds 50 per cent of the market share.

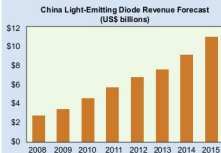
Taiwan, the global production base of consumer electronics products, is mainly engaged in the production of LEDs that emit visible light. Taiwan is also the world's largest downstream packaging and middle-stream chip production base.

With more than 30 years in LED development, China has formed a complete industrial chain in LEDs, covering the LED, epitaxial wafer, chip package and application. At present, China has over 1100 LED enterprises specialising in downstream packaging and application. But the development of the epitaxial wafer and chip in China is slow. There are only about ten enterprises engaged in the production of epitaxial wafer for LED, other than a few chip manufacturers.

China's LED industry gallops to Rs 250 trillion in 2011

Taking full advantage of government support, China's LED industry clocked sales of over Rs 250 trillion in 2011—23 per cent more than in 2010—by offering a wide range of innovative products. The industry is poised to grow almost double to Rs 460 trillion by 2015.

The LED market covers LED bulbs, LED displays, traffic signals, automotive applications, LCD backlighting, handset keypads, digital still camera flashlights, decorative lighting, street lighting and general illumination. A new demand driver for LEDs in the medium to long term will be the general lighting market. Given the global trend to reduce carbon emissions, China's



demand in the general lighting segment will be strong for LEDs, which offer low power consumption and are environmentally safe. The Chinese, however, are not happy since most LED wafers are imported. China trails due to the lack of technological and R&D capabilities. China's LED players enjoy ample funding from the government, which subsidises at least 70 per cent of the purchase price for metal-organic chemical vapour deposition equipment employed in LED manufacturing. Furthermore, tax and utility payment benefits are offered to encourage investments in the domestic LED industry, proving to be an additional boon for local players.

Source: A press release by IHS iSuppli China Electronics Supply Chain Service

In 2010, many patents in the LED industry expired. Chinese enterprises took advantage of this and broke open the shackles of intellectual property rights of European, American and Japanese giants. It is believed that they will make good use of huge market bases and abundant labour resources in order to occupy a place in the global LED market.

Indian LED industry

Despite the availability of cheaper technical manpower than China, India does little in manufacturing of LEDs; it does nothing at the mono-crystalline chip level and also nothing at the level of packaging chips into sources for lighting in various forms. Very small application level does exist, but it is mostly based on kits from China. Indian LED sources are just traders of mostly Chinese end products.

The government of India seems to be lacking concrete plans to develop and grow the LED industry in the

country. The investor community too seems to be following suit and investing unwisely in downstream levels of solar PV. With so much haphazard investment, many bankruptcies are likely to happen in the solar PV sector. Even the PV cell manufacturers in India are trading products from China and Taiwan. China too will see an even more serious crash.

Out of India's total electrical power, over 60 per cent is used in domestic and commercial areas for lighting. The prices of LEDs will continue to fall as high initial investments get depreciated and competition drives down prices. Today, in LEDs, there is a 70 per cent value addition at the chip level. In the next ten years, one might see LED prices as low as, or cheaper than, conventional light sources. ●

The author was formerly chairman of Electronics Commission of the government of India, and electronics advisor to India's late prime minister Rajiv Gandhi



Can I perform at Electronics Rocks?

Find out on Page No. 99 or visit www.electronic rocks.com

EFY GROUP
INDIA